

# Hemant Kumar Mahto

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## PROFESSIONAL EXPERIENCE

**Futureense Technologies**, *Tech Intern (3 months) active*

07/2026 – 10/2026 | Bengaluru

- **Automated the end-to-end AI interview question pipeline**—research, web extraction, LLM synthesis, RAG, question/answer generation, validation, deduplication, and persistence—to improve question quality for **1,000+ students**.
- **Redesigned and enhanced the Python/FastAPI backend**, integrating the automated pipeline with PostgreSQL, RAG services, APIs, authentication, configuration, and existing interview workflows.
- **Built automated quality-control and reliability mechanisms** including research freshness, relevance/contamination checks, answer validation, semantic deduplication, fail-closed validation, retries, and cost/quota controls.
- **Automated role- and company-specific knowledge processing**, enabling the system to continuously research, update, version, and reuse interview knowledge through the RAG pipeline instead of relying on manual question creation.
- **Improved backend scalability and maintainability** by separating research, synthesis, retrieval, generation, validation, and persistence into modular services and automating their orchestration end-to-end.

**366PI** , *Web Application Intern (6 weeks)*

- Worked on enterprise web and collaboration technologies, primarily **SharePoint Online**, gaining practical exposure to **SPFx, React, ASP.NET Core, Azure, Power BI, and DevOps** concepts.
- Implemented and supported **HotSync functionality to synchronize SharePoint Lists with Microsoft SQL Server**, enabling reliable data flow between SharePoint and backend systems.
- Configured **Azure-based authentication using certificates** to securely connect SharePoint Online with MS SQL, ensuring secure and automated data synchronization.
- **Deployed the Windmill Automation App review to the company's staging server**, validating functionality, configurations, and deployment readiness.

## PROJECTS

**E-Commerce Backend Application** 

**Tech Stack:** Java, Spring Boot, Spring Security, JWT, Spring Data JPA, Hibernate, MySQL, Maven, Lombok, Postman

- Developed a scalable e-commerce backend application using Spring Boot following layered architecture (Controller, Service, Repository).
- Implemented JWT-based authentication and role-based authorization using Spring Security.
- Built REST APIs for product catalog, category, shopping cart, address, order, and payment management.
- Designed database entities and relationships for users, products, carts, orders, payments, and addresses using JPA and Hibernate.
- Applied DTO pattern, validation, and global exception handling to improve security and maintainability.
- Integrated MySQL database using Spring Data JPA and Hibernate ORM for efficient data persistence.
- Tested and documented APIs using Postman while following RESTful API design principles.

**Intelligent Decision System using Reinforcement Learning and RAG** 

**Tech Stack:** React, Flask, Reinforcement Learning, FAISS, Gemini API

- Developed a Reinforcement Learning (Q-learning) based decision system using epsilon-greedy policy
- Integrated RAG using FAISS for contextual knowledge retrieval
- Added LLM-based reasoning (Gemini API) to enhance RL decision-making
- Built interactive UI to visualize agent actions and compare decision modules
- Deployed using Docker

## SKILLS

- Backend: Java, Spring Framework, Spring Boot, REST APIs, Spring Security, FAST API, Python 3
- Programming & CS Fundamentals: Data Structures & Algorithms
- Databases: MySQL, PostgreSQL
- Mobile Development: Flutter, Kotlin, Android Studio
- Frontend: React, HTML, CSS, JavaScript
- Generative AI & LLMs: RAG Pipelines, Gemini API, Vector Embeddings, pgvector, FAISS, Prompt Engineering
- ML & Deep Learning Fundamentals: Neural Networks, CNNs, RNNs, Transformers & Attention, Reinforcement Learning

## EDUCATION

**Master in Computer Application**, Amity University, Jharkhand  
(8.6 CGPA)

09/2024 – 07/2026 | Ranchi, Jharkhand

**Bachelor of Computer Applications (BCA)**, Marwari College, Ranchi, Jharkhand  
(8.40 CGPA)

2021 – 2024 | Ranchi, Jharkhand